

Forecasting Term Structure Dynamics with Multi-Output Gaussian Process Regression

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August 19th, 2025

Abstract

Term structure forecasting is important for both investment strategy and public policy. In this paper, we implement a non-parametric Bayesian framework to forecast yield curve dynamics using multi-output Gaussian process regression. We test multiple model specifications to identify suitable kernel structures for yield curve modeling and compare accuracy with traditional forecasting methods. In particular, we evaluate forecasts probabilistically using a log-scoring rule, assessing both point estimates and uncertainty. Our results show that the Gaussian process approach can improve out-of-sample performance and provides meaningful quantification of predictive uncertainty. We also develop a way for past volatility to affect forecast uncertainty by using an exponentially weighted moving average. We find this allows us to assign higher uncertainty in volatile periods, improving performance in log-score.

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1 Introduction

The forecasting of term structures is a well-studied problem due to its importance in guiding both investment strategy and public policy. The "yield curve", otherwise known as the term structure of interest rates, describes the relationship between different bonds' rates of return and their respective times to maturity. As a macroeconomic indicator, an upward sloping yield curve, where longer maturities have higher yields than shorter maturities, can suggest investors are optimistic about the future. When investors expect economic growth and its associated inflation, they demand higher compensation for longer maturities. Conversely, inverted yield curves tend to "predict" recessions; downturns in output have often followed periods when short-maturity yields have exceeded long-maturity yields.

Correctly predicting term structure dynamics can benefit investors both through directly profiting on bond trading and as a macroeconomic signal to adjust the allocations of other assets. For the purposes of central banks, yield curves can help to gauge markets' risk appetite and confidence, therefore guiding monetary policy and its consequences. Meanwhile, governments use them to manage debt allocations and inform fiscal policy as treasuries would like to know how much borrowing will cost them in the future.

A Dynamic Term Structure Model (DTSM) is a statistical model concerned with how the yield curve changes over time. Although DTSMs are varied in their approach, they generally aim to model yields as a function of time and potentially other factors. As the selection of different maturities is vast, most DTSMs consider an appropriately spaced subset of maturities and then attempt to describe their dynamics using a smaller number of factors (e.g. principal components). An issue that many DTSMs face is that if they include macroeconomic variables, they assume that yields depend on them linearly. This assumption may be too restrictive to describe yield data and is relaxed when using Gaussian process regression. This approach assumes a general functional form and is optimized from a Bayesian perspective. While this is more computationally expensive and does not restrict arbitrage, it is also more flexible and provides a natural framework for quantifying uncertainty.

The aim of this study is to develop a framework to forecast term structure dynamics from a non-parametric Bayesian perspective using multi-output Gaussian process regression. Aiming to improve performance out-of-sample, we then test different specifications in order to identify a suitable structure for yield curve modelling and compare their accuracy with traditional ways of forecasting the term structure. In particular, we will assess models' forecasts from a probabilistic point of view using a scoring rule, thereby assessing both point estimates and their associated standard errors.

The remainder of this paper is structured as follows. Section 2 presents an overview of traditional ways to model the term structure and previous applications of Gaussian processes in time series analysis and other supervised learning settings. Section 3 details how Gaussian process regressions work and some popular covariance structures along with a set of candidate models and ways we evaluate forecasts. Section 4 describes and interprets our empirical results, discussing their implications and their limitations. Finally, Section 5 concludes our findings and suggests directions for future work.

2 Literature Review

2.1 Term Structure Models

A wide range of models have been proposed to describe the term structure. They often balance competing aims of economic intuition, empirical fit, and flexibility. The following review outlines common approaches and discusses their implications and limitations. We also highlight the trade-offs that motivate the use of a flexible, data-driven approach such as Gaussian process regression.

Nelson and Siegel (1987) develop an early method for effectively modelling term structures in a parsimonious way. With only four parameters to estimate, they propose a simple way to fit the term structure that remains supported empirically. They initially propose an expression for the theoretical interest rate over an infinitesimal period called the instantaneous forward rate. By varying the maturity, they create the forward curve which they then integrate to derive the yield curve. Their formulation is both parsimonious and interpretable, with the four parameters representing the level, slope, curvature of the yield curve and the decay rate which determines the location of the hump. These characteristics are common ways to describe the term structure which form the basis for how later models approach this task. However, each fitted model contains only a static representation of the yield curve based on current data and does not attempt to describe how the term structure changes over time.

$$R(m) = \beta_0 + (\beta_1 + \beta_2)\frac{1 - exp(-\frac{m}{\tau})}{\frac{m}{\tau}} - \beta_2 exp(-\frac{m}{\tau}) \tag{1}$$

Developing this idea into a dynamic model, Diebold and Li (2006) use Nelson & Siegel's structural form but treat the parameters β_0, β_1, and β_2 as time-varying latent factors. By estimating these parameters cross-sectionally (that is, at each individual time period), they obtain time series for β_0, β_1 and β_2 which they can then model using a vector auto-regressive (VAR) process. This approach maintains the parsimonious nature of Nelson & Siegel while now allowing for the forecasting of the term structure. While this model has been shown to perform well in and out-of-sample, their approach fixes the decay parameter λ, meaning it cannot detect changes in the location of the hump and does not make any attempt to restrict arbitrage. Arbitrage, simply speaking, involves simultaneously buying and selling an asset to take advantage of a price difference. The lack of arbitrage is a common assumption in financial models as we should expect any such opportunity to be immediately bought up. The modelling of how the level, slope and curvature change over time is an approach that is shared by later factor models, which we will use as a benchmark for comparison to the Gaussian process framework.

Another prominent method is the Affine Term Structure Model (ATSM) as described by Nelson and Siegel (1987). They set out a model where all yields are some affine (linear) combination of state variables, X_t. That is, the short rate, r_t, follows $r_t = d_0 + d_1 X_t$ and

the state vector X_t follows a stochastic process called an Ornstein–Uhlenbeck (OU) process. With an arbitrage-free argument, they show that both the logarithm of the bond price and the yield to maturity are affine in the state vector if the derived Riccati differential equations are tractably solvable, which is satisfied in the Gaussian case. This affine class of term structure models has been adapted various times for different purposes. Ang and Piazzesi (2003) incorporate macroeconomic variables jointly with yield state variables in a VAR formulation and Christensen et al. (2011) bridge the gap between the affine and Nelson–Siegel approach by keeping the interpretable structural form but adding an extra term to ensure no arbitrage. However, these models are complex to estimate and lack the parsimony of Nelson–Siegel formulations which naturally hurts performance out-of-sample. In fact, Duffee (2002) shows that completely affine models can forecast worse than a random walk benchmark and diagnoses the problem as an overly restrictive link between risk premia and volatility. Enforcing no arbitrage by construction does incorporate economic intuition into the model but may also be too restrictive to remain supported empirically. This motivates a shift towards observable factor models and principal components which attempt to balance intuition with predictive accuracy.

The previous models use latent factors in their characterisation of the yield curve. More specifically, they assume that the yields follow latent factors which evolve over time. In this framework, imposing the no-arbitrage condition influences the statistical inference. Conversely, Joslin et al. (2011) prove that in any canonical Gaussian dynamic term structure model with observed factors (of the yields), a no arbitrage restriction under the physical measure will not impact the maximum likelihood estimates of an unrestricted VAR and therefore cannot impact forecasts. Their proposal for observable factors of yield curve principal components (PCs) allows for easy and parsimonious estimation and forecasting. Litterman and Scheinkman (1991) show that only three factors are needed to describe almost all variation in treasury bond yields. While this approach is simple and data-driven, there is scope for more flexibility in allowing factors to evolve non-linearly over time.

Next, researchers question whether those principal components are enough. There is discussion in the literature about whether the yield curve spans the macroeconomy, that is, if values of historical treasury yields contain all relevant information necessary to forecast future yields. If so, macroeconomic indicators will not improve our forecasts. Ludvigson and Ng (2009) use principal component analysis on a large panel of 132 macroeconomic indicator series to extract 9 PCs to measure the macroeconomy. They argue including factors that measure economic activity, inflation and the Cochrane and Piazzesi (2005) forward rate factor predicts yields more accurately than without macroeconomic variables.

Across these approaches, we can see that while ATSMs successfully restrict arbitrage, they can be too complex in estimation to generalise out-of-sample. Furthermore, macroeconomic and observed factor models are parsimonious and data-driven but lack the ability to capture non-linear yield curve dynamics. These limitations motivate the use of multi-output Gaussian processes. This approach still models yields as linear combinations of fewer factors but allows them to evolve non-linearly over time and produces probabilistic predictions.

2.2 Gaussian Process Regression

A stochastic process is said to be a Gaussian process (GP) if any finite subset of the process follows a multivariate Normal distribution. Gaussian process regressions (GPR) are non-parametric Bayesian models which allow for a general functional form by assuming that f follows a GP *a priori*.

$$y_i = f(x_i) + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2) \tag{2}$$

The prior is fully specified once we choose a covariance function, also known as a kernel. This governs the covariance that any two points share and determines the shape of the function. Under Gaussian noise, the posterior and predictive distributions can be derived analytically, lending the benefit of tractable exact inference and probabilistic prediction.

GPR was first used and developed under the name of kriging (Matheron (1960)) as a method for interpolating spatial data. Its flexibility and natural way of quantifying uncertainty has led to its widespread use in this field. Introduced in the machine learning literature by Rasmussen and Williams (2005), GPR has become a powerful tool for non-linear regression and classification tasks.

GPR also extends to supervised learning problems of multiple dimensions. As detailed in Alvarez et al. (2012), the intrinsic model of coregionalization, the semi-parametric latent factor model and, more generally, the linear model of coregionalization each offer related specifications for modelling multiple correlated outputs and is described in more detail in Section 3.

As an alternative to traditional time series methods such as the autoregressive integrated moving average (ARIMA) or vector auto-regressive (VAR) models, GPR has been proposed as a non-linear way to describe and forecast time series data (Roberts et al. (2013)). Viewed in continuous time, GPR handles missing or unevenly spaced data easily as its inference and prediction are unchanged. While many popular kernels are stationary, non-constant variance can be addressed using heteroscedastic GPs (Le et al. (2005)), though exact inference is no longer tractable.

A notable limitation of GPR is its computational complexity; its time complexity scales in $\mathcal{O}(n^3)$ where n is the training size. As such, attempts have been made to curb this cubic scaling. In Snelson and Ghahramani (2005), $m \ll n$ inducing points — that is, points that infer the shape of the function — are used and learned by maximizing the marginal likelihood alongside other hyperparameters. This reduces complexity to $\mathcal{O}(nm^2)$. The current state-of-the-art follows Titsias (2009), where inducing point locations are learned with hyperparameters through variational inference to avoid overfitting. This is a principled way to approximate the exact function by minimizing a lower bound on the marginal likelihood.

Despite its complexity, GPR has potential for effective yield curve modelling. This study joins the few that use GPR and proposes a unified way to model term structures.

3 Methodology

3.1 Data

Our data includes monthly yields for US Treasury bonds from January 1988 to January 2020. The chosen maturities are 1, 2, 3, 4, 5, 7 and 10 year bonds to cover a wide enough range of the yield curve while avoiding noisy short-term maturities under one year. Thus, we measure time in years. Figure 1 shows the movements of treasury yields over this period. We observe a general decline in level over the years and periods of a steeper slope between 1990–1995, 2000–2005 and 2007–2018.

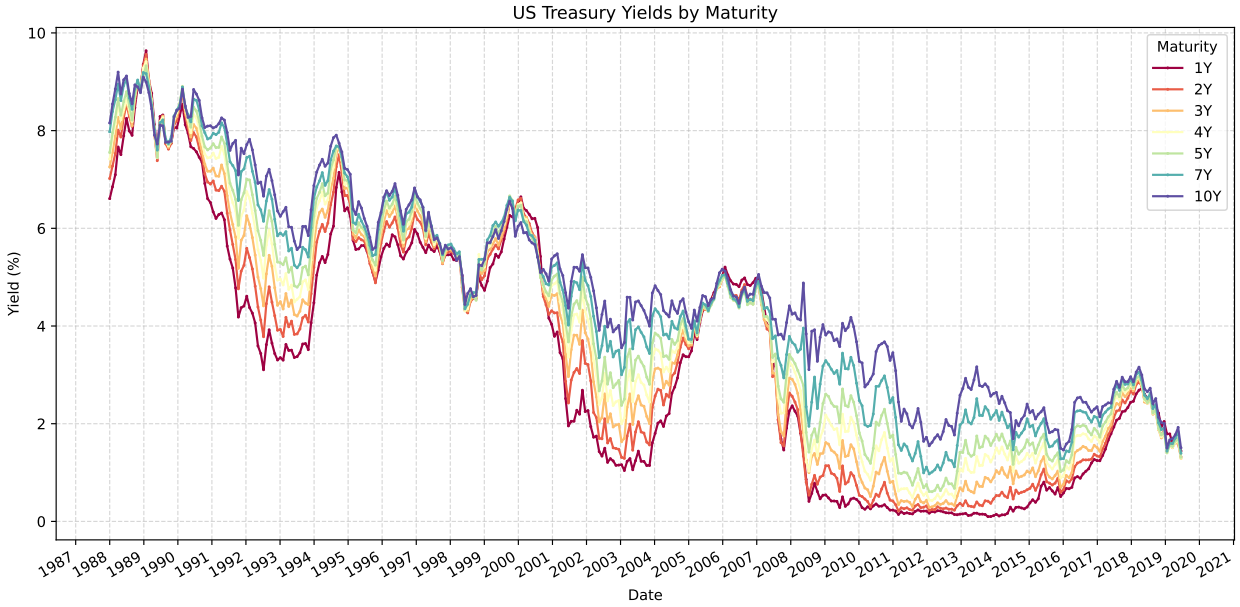


Figure 1: Movement of Treasury yields over time.

From the Federal Reserve Bank of St. Louis Database (FRED), we have data for year-on-year CPI inflation and the federal funds effective rate (U.S. Bureau of Labor Statistics (2025), Federal Reserve Bank of St. Louis (2025)). Generally, both CPI inflation and federal funds rate move with treasury yields (2). However, as it is determined by the Federal Open Market Committee, the federal funds rate is much more reactive than CPI inflation which, being a 12-month rolling figure, moves more gradually.

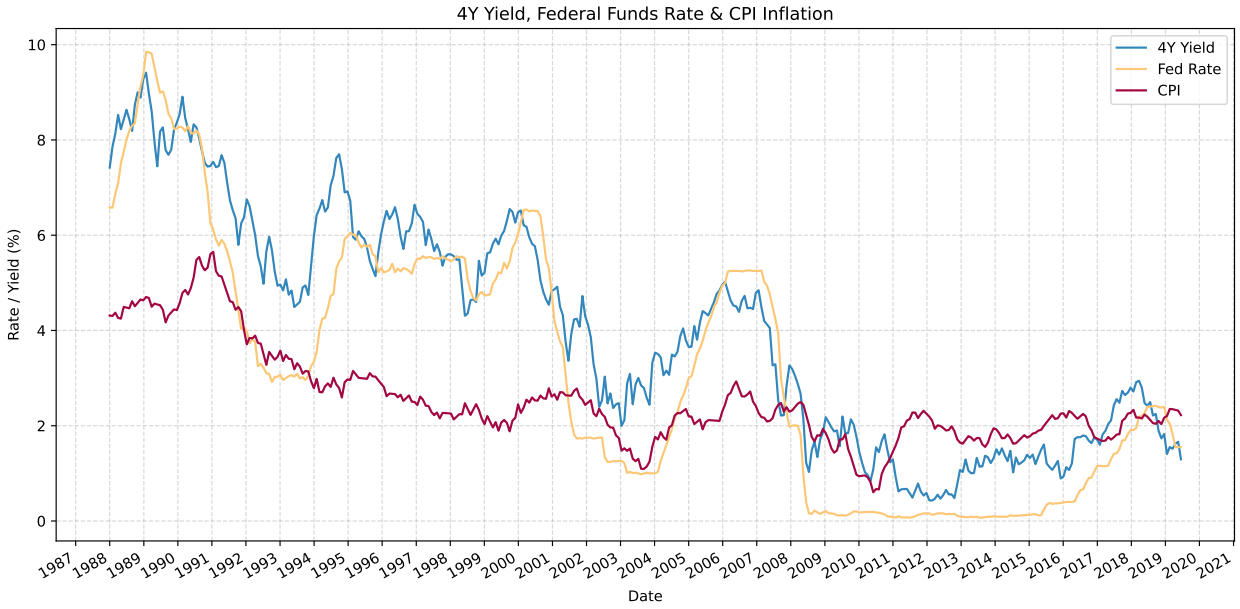


Figure 2: Treasury yields, federal funds rate, and CPI inflation over time.

Additionally, we use the nine macroeconomic factors from Ludvigson and Ng (2009) to describe the broad dynamics of the macroeconomy across the same period. These are standardised principal components of 132 time series to measure broad macroeconomic conditions. These factors mostly behave like white noise except for a large spike around the global financial crisis of 2008.

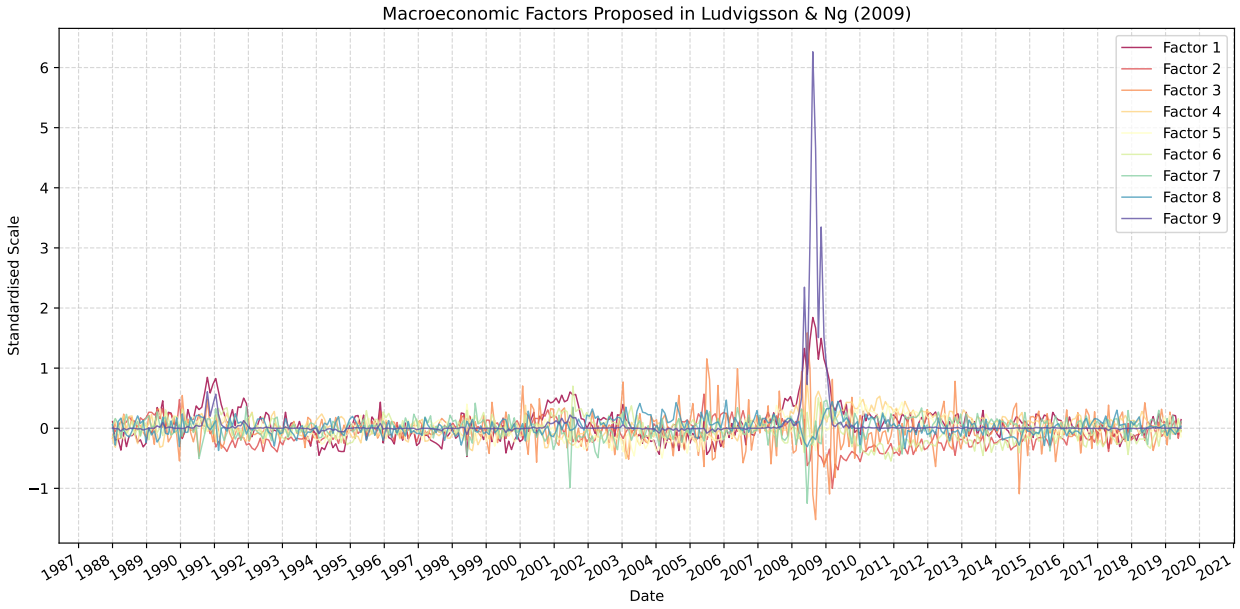


Figure 3: Macroeconomic factor series (principal components).

Finally, we construct a series to represent past volatility of our treasury yields by taking an exponentially weighted moving average of first-differenced yields. First, take $\Delta Y_t = Y_t - Y_{t-1}$ and initialise a covariance matrix between all maturities using historical yields up to January 1988. Then recursively set

$$\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) \Delta Y_t \Delta Y_t^\top \tag{3}$$

for each month onwards where λ is the decay parameter that controls how much past volatility affects the present value. This creates a volatility tensor of size $7 \times 7 \times n$ of which we take the trace of each 7×7 matrix. This gives us an indicator for past volatility at each time point by using the sum of the marginal volatilities across each maturity. We use $\lambda = 0.95$, a popular value in risk management which maintains a smooth level of past volatility, does not exaggerate recent noise and keeps an effective memory length of around 25 months.

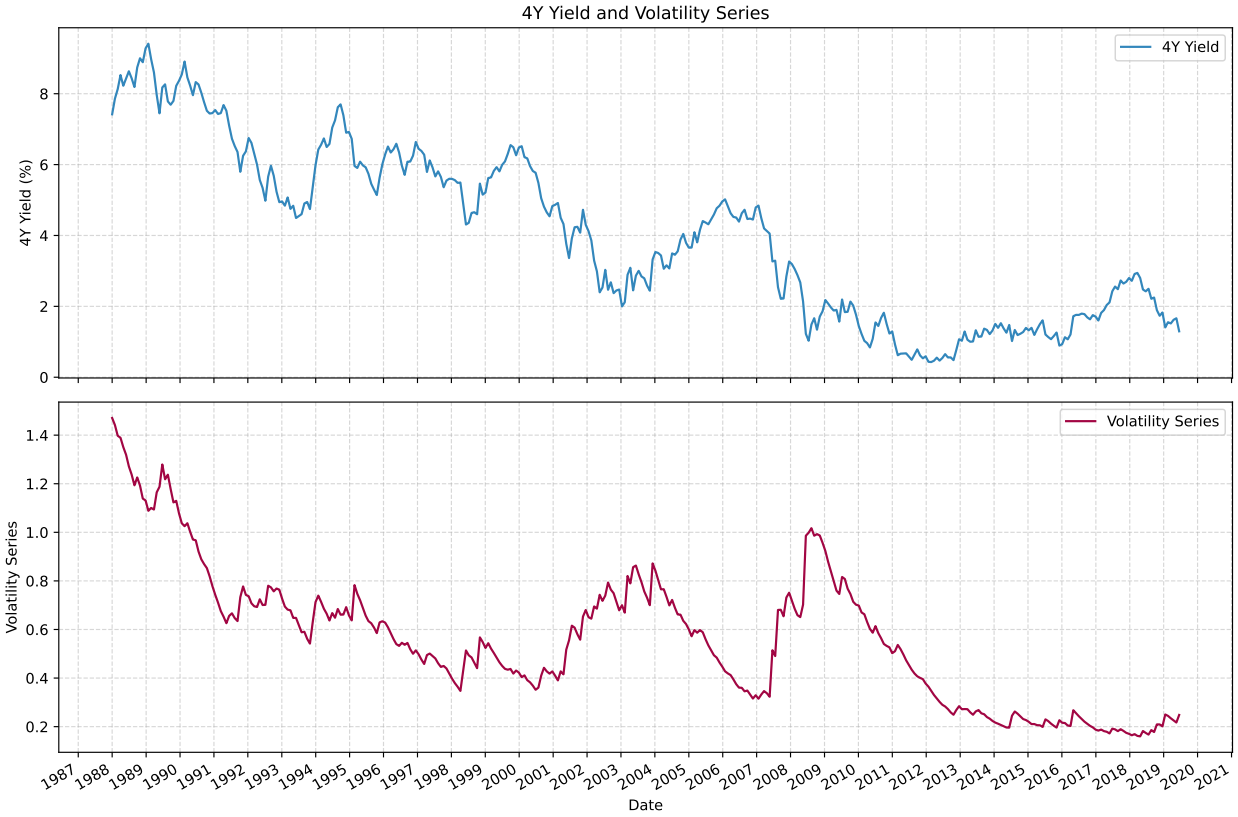


Figure 4: Treasury yields and derived volatility series.

3.2 Multi-Output Gaussian Process Regression

Now consider a vector of target variables $(Y_1, \ldots, Y_D)^\top$ and a vector of features $(X_1, \ldots, X_p)$ for which we can describe a supervised learning task of the form

$$\mathbf{Y} = \mathbf{f}(\mathbf{X}) + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma). \tag{4}$$

where $\Sigma \in \mathbb{R}^{D \times D}$ is a diagonal matrix of independent noise parameters $\sigma_i^2, i = 1, \ldots, D$.

For our purposes, we use the Linear Model of Coregionalization (LMC) as described in Alvarez et al. (2012). This will allow us to model cross-correlations between treasury yields of different maturities and offers a built-in method of dimension reduction that helps avoid overfitting.

The LMC says that each output $d = 1, \ldots, D$ is a linear combination of Q different kernel structures, each with R_q independent latent functions:

$$f_d(\mathbf{X}) = \sum_{q=1}^{Q} \sum_{i=1}^{R_q} a_{d,q}^i u_q^i(\mathbf{X}). \tag{5}$$

Here a_{dq} are mixing weights between the latent functions. We place a GP prior on these functions such that each $u_q^i(\mathbf{x}) \sim \mathcal{GP}(0, k_q(\mathbf{x}, \mathbf{x}'))$. By the formulation of the LMC, the covariance between any two outputs d and d' is therefore

$$\text{cov}[f_d(\mathbf{x}), f_{d'}(\mathbf{x}')] = \sum_{q=1}^{Q} \sum_{q'=1}^{Q} \sum_{i=1}^{R_q} \sum_{i'=1}^{R_q} a_{d,q}^i \, a_{d',q'}^{i'} \text{cov}[u_q^i(\mathbf{x}), u_{q'}^{i'}(\mathbf{x}')]. \tag{6}$$

This is the full covariance between all outputs and inputs and is equivalently written as

$$\mathbf{K}(\mathbf{x}, \mathbf{x}') = \sum_{q=1}^{Q} \mathbf{B}_q k_q(\mathbf{x}, \mathbf{x}'). \tag{7}$$

Each $\mathbf{B}_q$ is known as a coregionalization matrix and contains the mixing weights for each latent function in the kernel structure. It governs the correlations between each output and has rank equal to the number of latent functions, R_q. Using $R_q \ll D$, we have a built-in method of dimension reduction analogous to that of the VAR on PCs approach. This allows us to adapt our model to the complexity of our problem and our data so that we can avoid both over- and under-fitting. If we have $Q = 1$ different kernel structures, the LMC simplifies to the Intrinsic Model of Coregionalization (IMC) where the outputs are linear combinations of R_q independent functions, each with the same prior.

3.2.1 Deriving the Posterior and Predictive Distributions

Under our model's prior assumptions, $\mathbf{Y} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma})$, where $\Sigma \in \mathbb{R}^{D \times D}$ is a diagonal matrix of independent noise parameters $\sigma_i^2, i = 1, \ldots, D$. we have the joint distribution of $\mathbf{Y}$ and the functions for our test data, $\mathbf{f}_*$, as

$$\begin{pmatrix} \mathbf{Y} \\ \mathbf{f}_* \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma} & \mathbf{K}(\mathbf{X}, \mathbf{X}_*) \\ \mathbf{K}(\mathbf{X}_*, \mathbf{X}) & \mathbf{K}(\mathbf{X}_*, \mathbf{X}_*) \end{pmatrix}\right).$$

where $\mathbf{K}(\mathbf{X}, \mathbf{X}_*)$ is the covariance between train and test inputs. By properties of multivariate Normal distributions, we have that

$$\mathbf{f}_*|\mathbf{Y}, \mathbf{X}, \mathbf{X}_* \sim \mathcal{N}(\mu_*, \mathbf{\Sigma}_*)$$

where

$$\mu_* = \mathbf{K_{x_*}}^\top (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma})^{-1} \bar{\mathbf{y}},$$
$$\mathbf{\Sigma}_* = \mathbf{K}(\mathbf{X}_*, \mathbf{X}_*) - \mathbf{K_{x_*}} (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma})^{-1} \mathbf{K_{x_*}}^\top.$$

So it follows that as $\mathbf{Y}_* = \mathbf{f}_*(\mathbf{X}_*) + \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma)$,

$$\mathbf{Y}_*|\mathbf{Y}, \mathbf{X}, \mathbf{f}_*, \mathbf{X}_* \sim \mathcal{N}(\mu_*, \mathbf{\Sigma}_* + \mathbf{\Sigma}).$$

3.2.2 Estimation of Kernel Parameters

As per Alvarez et al. (2012), we often go about estimating the parameters of our Gaussian process regression, $\boldsymbol{\phi}$, through maximising the marginal likelihood, $p(\mathbf{Y} \mid \mathbf{X}, \boldsymbol{\phi})$. Given a Gaussian likelihood, $\mathbf{f}$ can be integrated out to leave

$$p(\mathbf{Y} \mid \mathbf{X}, \boldsymbol{\phi}) \sim \mathcal{N}(\mathbf{Y} \mid 0, \mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma}), \tag{8}$$

meaning the log-marginal likelihood can be written

$$\log p(\mathbf{Y} \mid \mathbf{X}, \boldsymbol{\phi}) = -\tfrac{1}{2}\mathbf{Y}^T (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma})^{-1} \mathbf{Y} - \tfrac{1}{2} \log |\mathbf{K}(\mathbf{X}, \mathbf{X}) + \mathbf{\Sigma})| - \tfrac{nD}{2} \log 2\pi. \tag{9}$$

This is our objective function and is maximised with respect to each parameter in $\boldsymbol{\phi}$ with a gradient-based method. We interpret the first term of this expression as encouraging goodness-of-fit while the second term penalises complexity, demonstrating the innate regularisation of maximising the evidence in a GPR.

3.3 Kernel Choice

Choice of kernel is crucial for any Gaussian process regression as it governs the covariance between inputs. For the purposes of time series analysis, the central task is picking a kernel that fits the temporal structure of our series. Such a kernel adequately describes and accounts for shocks and seasonality while matching the smoothness of the data. We now outline a few common non-stationary and stationary kernels and describe their uses. Recall that a stochastic process is (weakly) stationary if and only if its mean and covariance are finite and depend only on the distance between two points, rather than their absolute values.

3.3.1 Non-stationary Kernels

Dot Product Kernel

The Dot Product kernel represents a linear relationship between inputs. It is defined as

$$k(x, x') = \sigma^2(x^T x' + c)^d, \tag{10}$$

where σ^2 is the variance, c is the bias parameter and d is the order parameter. When $d = 1$, this is the linear kernel and this choice of covariance function is equivalent to performing Bayesian Linear Regression (Box and Tiao (1973)).

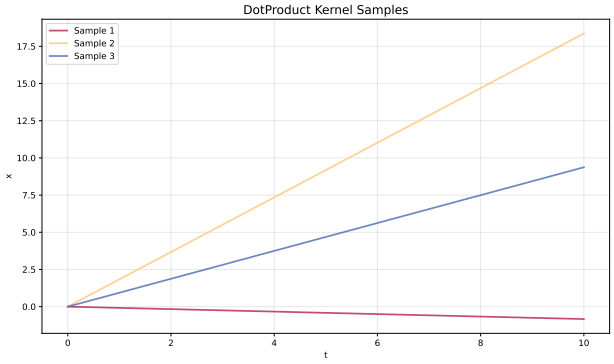


Figure 5: Three samples using the DotProduct kernel.

Brownian Kernel

The Brownian kernel describes the covariance of a Brownian (Wiener) process and is defined as

$$k(x, x') = \sigma^2 \min(x, x'), \tag{11}$$

where σ^2 is the variance of the process. The resulting function is continuous but not differentiable. As a result, the kernel can be useful for modelling rough series that show random walk-like autocorrelation.

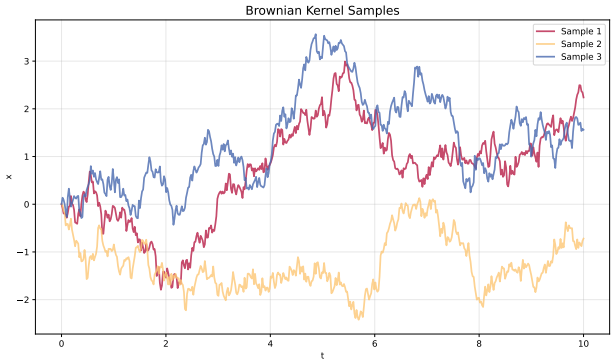


Figure 6: Three samples using the Brownian motion kernel.

3.3.2 Stationary Kernels

Ornstein–Uhlenbeck (Exponential) Kernel

The Exponential, or OU, kernel describes the covariance of an OU process. OU processes are Gaussian and Markov processes; setting this prior for our latent functions can be useful for describing data that is rough and mean-reverting. It is important to note that while the prior can be a Markov process, no GP posterior or predictive distribution can be Markov as it always depends on every training point. The formulation of the OU is as such

$$k(x, x') = \sigma^2 \exp\left(-\frac{|x-x'|}{\ell}\right), \tag{8}$$

where the lengthscale parameter ℓ controls the decay in covariance between points and the variance parameter σ^2 controls the variance of the process. It can be viewed as a continuous time analogue of the AR(1) process and is also equivalent to the more general Matern kernel with $\nu = 1/2$:

$$k(x, x') = \sigma^2 \cdot \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}|x-x'|}{\ell}\right)^{\nu} K_\nu \left(\frac{\sqrt{2\nu}|x-x'|}{\ell}\right) \tag{9}$$

where $K_\nu(\cdot)$ here is a Bessel function. This has the same lengthscale and variance parameters but the extra parameter ν controls the smoothness of the function. This structure is especially useful for data where we have prior beliefs about the differentiability of our process and different values for ν offer a good middle ground between the roughness and smoothness of the OU and RBF structures.

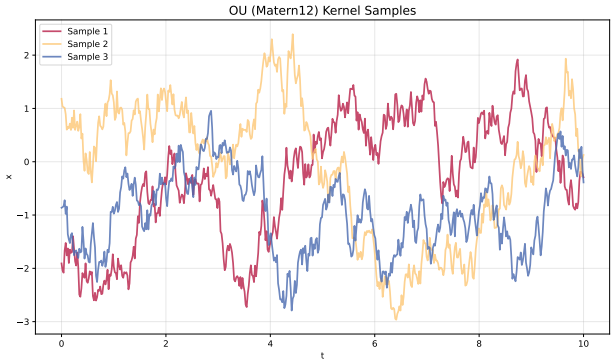


Figure 7: Three samples using the OU kernel.

Squared Exponential (RBF) Kernel

The Radial Basis Function (RBF) or Squared Exponential kernel is widely used in Gaussian process modelling due to its flexibility. It can be seen as a Matern kernel where $\nu \to \infty$ and it produces smooth functions that are mean-reverting and infinitely differentiable. The ℓ and σ^2 parameters again control the decay and variance of the process and the kernel is defined as

$$k(x, x') = \sigma^2 \exp\left(-\frac{(x-x')^2}{2\ell^2}\right). \tag{10}$$

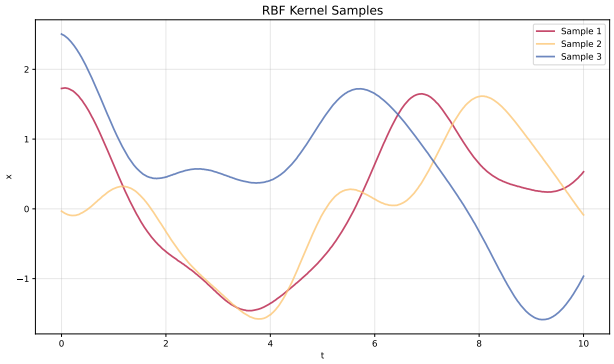


Figure 8: Three samples using the RBF kernel.

Standard Periodic Kernel

The standard periodic kernel allows us to model seasonality in our data by producing smooth oscillations for a fixed period. Again, it contains lengthscale and variance parameters but now has a period p, and is defined as

$$k(x, x') = \sigma^2 \exp\left(-\frac{2\sin^2(\pi|x-x'|/p)}{\ell^2}\right). \tag{11}$$

This structure is limited in that it produces identical oscillations on a fixed period. However, when combined with an RBF kernel, it can produce functions that contain some seasonality but are allowed to modulate which can be useful for modelling trends in time series data.

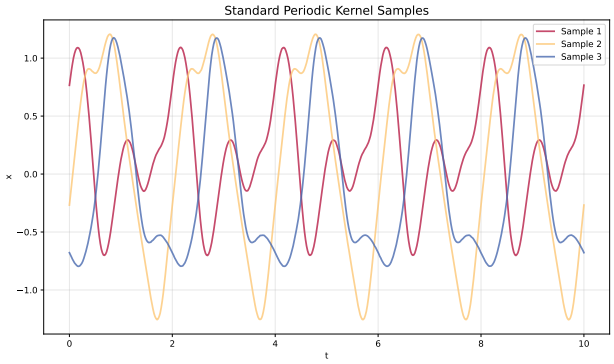


Figure 9: Three samples using the Standard Periodic kernel.

Other structures are available such as the polynomial, spectral mixture kernel and more. See Rasmussen and Williams (2005) for more details.

Combining Kernels

Recall a covariance function, $k(x, x')$, is valid if and only if it is symmetric and positive semi-definite. Therefore, the sum or product of any two kernels creates a new valid kernel function. The sum of two kernels is often thought of as an "or" operation and corresponds to adding two independent GPs with each structure. Conversely, the product of two kernels is thought of as an "and" operation or an interaction between components of the covariance structure. For example, the popular RBF multiplied by Standard Period kernel creates a new "locally periodic" kernel that exhibits seasonality in some areas but has the flexibility to drift in and out of strict periodicity. Using these combinations, we can create expressive custom covariance structures that may better fit our data.

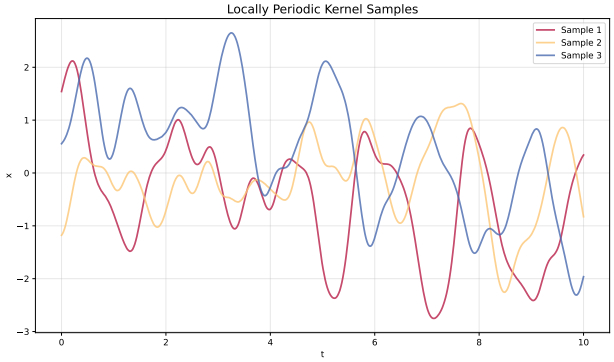


Figure 10: Three samples using the 'locally periodic' kernel.

3.4 Candidate Models

3.4.1 Gaussian Process Models

We now outline a set of candidate models used in this study. These will then be compared to benchmark models described in 3.4.2.

Model 1A – Brownian Kernel IMC

First, we propose an IMC of 3 functions that follow a Brownian motion, *a priori*. Let $\mathbf{Y}$ be a vector of 1,2,3,4,5,7 and 10 year treasury yields and $\mathbf{X}$ be time in years. $\mathbf{f}(\mathbf{X})$ is a vector $f_1(\mathbf{X}), \ldots, f_7(\mathbf{X})$.

$$\mathbf{Y}_t = \mathbf{f}(\mathbf{X}_t) + \boldsymbol{\epsilon}_t, \quad f_d(\mathbf{X_t}) = \sum_{i=1}^{R_q} a_d^i u^i(\mathbf{X}_t), \quad u_i(x) \sim \mathcal{GP}(0, k(x, x')) \tag{12}$$

$$k(x, x') = \sigma^2 \min(x, x'), \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma). \tag{13}$$

Here we are expressing each maturity as a linear combination of three latent functions. This demonstrates the built-in dimension reduction of multi-output Gaussian process regression as we typically choose less latent functions than dimensions, exploiting the shared dynamics of our series. Under this Brownian framework, the latent factor priors are assumed to be Markov without any mean-reversion and can be seen as a continuous equivalent of a random walk with variance growing linearly over time. It is important to note that the functions are no longer Markov *a posteriori* because they have been conditioned on all past data, information which is used to inform both the posterior mean and covariance. Being a special case of the LMC with $Q = 1$ groups of kernel structures, the IMC offers a simplification that may be useful given the strong association between yields of each maturity. This simpler model may often be sufficient to describe the shared dynamics of each output and can prevent a model from overfitting, improving its performance out-of-sample. Practically speaking, we identify the optimal number of latent functions, R_q, through cross-validation of each model's marginal likelihood.

Model 1B – OU Kernel IMC.

Treasury yields are generally thought to be mean-reverting (Vasicek 1977). We can make a slight adaption to Model 1A to encode this behaviour. Instead of using a Brownian kernel, we model the prior functions as OU processes such that

$$k(x, x') = \sigma^2 \exp\left(-\tfrac{|x-x'|}{\ell}\right). \tag{14}$$

Every other aspect of the model is the same; however, the latent factor priors are now assumed to be mean-reverting and stationary, a continuous version of an AR(1) rather than a random walk. In fact, for a stationary AR(1) process with parameter ϕ,

$$Y_t = \phi Y_{t-1} + e_t, \quad e_t \sim \mathcal{N}(0, \sigma^2), \quad |\phi| < 1, \tag{15}$$

letting $\Delta t \to 0$, the AR process becomes an OU process with $\phi = 1 - \theta\Delta t$ where $\theta = 1/\ell$, the lengthscale of our kernel. For the case of the random walk where $\phi = 1$, we have that $\theta = 0$ meaning that as the lengthscale $\ell \to \infty$, the mean-reversion disappears and the process becomes a Brownian motion.

$$OUProcess : dx_t = -\theta x_t dt + \sigma dW_t$$

$$BrownianMotion : dx_t = \sigma dW_t$$

Model 1C – Locally Periodic + OU Kernel IMC

We can make our covariance function more expressive by adding together the OU kernel and the "locally periodic" RBF multiplied by Standard Periodic kernel. This formulation is again similar to models 1A and 1B but the covariance is now

$$k(x, x') = k_{OU}(x, x') + k_{RBF}(x, x')k_{Per}(x, x'). \tag{16}$$

Although this has more parameters to estimate, this is a more expressive kernel structure that allows our latent functions to drift smoothly in and out of periodicity while remaining rough and mean-reverting. Since the addition of two kernels can be equivalently viewed as the sum of two independent GPs, we can view this model as separately representing the trend with a smooth locally periodic function and the microscopic component with an OU process.

Model 1D – Time Kernel + Macro Kernel IMC

We extend this model further by adding lagged macroeconomic indicators through separate kernels. We keep the IMC of Model 1C but add a smooth RBF kernel to model the relation between each month's treasury yields and the CPI inflation and federal funds rate the month before. Now we have three dimensional $X = (X_t, X_{cpi}, X_{fed})$ and the covariance is

$$k(x, x') = k_{OU}(t, t') + k_{RBF}(t, t')k_{Per}(t, t') + k_{RBF}(X_{t-h,cpi}, X_{t-h,fed}). \tag{17}$$

where h is the number of lags ahead we intend to forecast. Here in the RBF kernel for the CPI and federal funds rate, we use Automatic Relevance Detection (ARD). By assigning each dimension its own lengthscale, the model learns how much weight to assign to each input, again aimed to limit overfitting. The lagged nature of the macroeconomic variables allows us to make predictions without having to forecast our inputs as well.

Model 1E – Including Volatility

A limitation of the OU, RBF and Standard Periodic kernels is that they are stationary and may not sufficiently describe the clustered volatility often prevalent in financial time series. By including the trace of an exponentially weighted moving average of past volatility between outputs, x_{vol}, we create a mechanism for past volatility to affect future predictions.

$$k(x, x') = k_{OU}(t, t') + k_{RBF}(t, t')k_{Per}(t, t') + k_{RBF}(X_{t-h,cpi}, X_{t-h,fed}) + k_{BM}(X_{t,vol}, X'_{t,vol}). \tag{18}$$

By including a Brownian kernel for x_{vol}, we introduce non-stationarity into our structure as variance is higher for larger values of past volatility. This allows the model to assign higher levels of uncertainty following recent volatility, thereby starting to account for volatility clustering.

Model 1F – More Macroeconomic Variables

For the final IMC, we use the nine macroeconomic factors from Ludvigsson & Ng (2009). This allows us to use almost all information in the macroeconomy to make one-step forecasts for the treasury yields. Containing all the macroeconomic variables inside one kernel, we will again use ARD for each input dimension so that $X = (X_t, X_{cpi}, X_{fed}, X_1, \ldots, X_9)$ and the covariance becomes

$$k(x, x') = k_{OU}(t, t') + k_{RBF}(t, t')k_{Per}(t, t') \tag{19}$$

$$+k_{RBF}(X_{t-h,cpi}, X_{t-h,fed}, X_{t-h,1}, \ldots, X_{t-h,9}) + k_{BM}(X_{t,vol}, X'_{t,vol}). \tag{20}$$

Having each of the macroeconomic variables and factors contained within one RBF kernel lets them interact in their effect on treasury yield dynamics.

Model 2A – OU & Trend LMC

In a linear model of coregionalization (LMC), the outputs are linear combinations of (possibly multiple) independent functions per kernel structure. We can form an LMC for treasury yields as such

$$\mathbf{Y}_t = \mathbf{f}(\mathbf{X}_t) + \boldsymbol{\epsilon}_t, \quad f_d(\mathbf{X}) = \sum_{q=1}^{2} a_{d,q} u_q(\mathbf{X}), \quad u_q(x) \sim \mathcal{GP}(0, k_q(x, x')) \tag{21}$$

$$k_1(x, x') = k_{OU}(t, t'), \quad k_2(x, x') = k_{RBF}(t, t')k_{Per}(t, t'), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma). \tag{22}$$

so that each $\mathbf{B}_q$ has rank 1. As the addition of kernels is equivalent to the summation of independent GPs, the only way this structure differs from Model 1C is that now each latent function has its own coregionalization matrix. Outputs therefore take different weights from the trend kernel than they do from the OU kernel.

Model 2B – CPI and Federal Funds LMC

Again, we extend the purely temporal model by adding a kernel for the lagged macroeconomic variables, CPI inflation and federal funds rate. The general form of the model is the same but now we have

$$f_d(\mathbf{X}) = \sum_{q=1}^{3} a_{d,q} u_q(\mathbf{X}), \quad u_q(x) \sim \mathcal{GP}(0, k_q(x, x')) \tag{23}$$

$$k_1(x, x') = k_{OU}(t, t'), \quad k_2(x, x') = k_{RBF}(t, t') k_{Per}(t, t'), \quad k_3(x, x') = k_{RBF}(X_{t-h,cpi}, X_{t-h,fed}). \tag{24}$$

Model 2C – More Macroeconomic Variables

Once again, we extend the macroeconomic kernel to include the Ludvigsson and Ng (2009) factors so that

$$k_3(x, x') = k_{RBF}(X_{t-h,cpi}, X_{t-h,fed}, X_{t-h,1}, \ldots, X_{t-h,9}). \tag{25}$$

Again, this lets the whole macroeconomy affect treasury yields jointly. However, its effect on yields is not necessarily parallel with time and makes this model more flexible.

3.4.2 Benchmark Models

The following section details the models that we compare the above GP regressions to with respect to their performance out-of-sample.

Model 3A – Random Walk

The most basic benchmark for any time series model is the random walk. This model is pessimistic in that past values cannot help forecast the future and each output is independent. It says

$$Y_{i,t} = Y_{i,t-1} + e_t, \quad e_t \sim \mathcal{N}(0, \sigma_i^2), \quad i = 1, \ldots, 7. \tag{26}$$

Here, any forecast is simply equal to the last observed point as

$$\mathbb{E}(Y_{i,t+l} \mid \mathcal{F}_t) = Y_{i,t} + \mathbb{E}\left(\sum_{j=1}^{l} e_i\right) = Y_{i,t}, \tag{27}$$

where $\mathcal{F}_t$ is the information up until time t. It follows that prediction variance grows linearly with l,

$$\mathrm{Var}_t(Y_{i,t+l}) = l \cdot \sigma_i^2. \tag{28}$$

Model 3B – ARIMA

The next benchmark we test against is the well-known auto-regressive integrated moving average (ARIMA) model. We again treat each output as independent so that the model is

$$\Phi_i(L)(1-L)^{d_i}Y_{i,t} = \Theta_i(L)\,\varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \mathcal{N}(0,\sigma_i^2), \quad i = 1,\ldots,7, \tag{29}$$

where $\Phi(\cdot)$ and $\Theta(\cdot)$ are the auto-regressive and moving average operators and L is the lag operator. We can choose hyperparameters either through inspection of the auto-correlation and partial auto-correlation function for each output or through some information criterion such as the Akaike Information Criterion (AIC) (Akaike (1973)).

Model 3C – VAR(1) on 3 PCs

As Joslin et al. (2011) say, arbitrage restrictions will not affect predictions under the physical measure for Gaussian ATSMs where the factors are observed. As such, we test our GP regressions against this specification where the observed factors are three PCs for the seven output dimensions modelled using a VAR(1) process:

$$\begin{bmatrix} Z_{1,t} \\ Z_{2,t} \\ Z_{3,t} \end{bmatrix} = \mathbf{\Phi} \begin{bmatrix} Z_{1,t-1} \\ Z_{2,t-1} \\ Z_{3,t-1} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \\ e_{3,t} \end{bmatrix}, \quad e_{i,t} \sim \mathcal{N}(0,\sigma_i^2). \tag{30}$$

where Z_1, Z_2, Z_3 are the first three principal components of our treasury yields and $\mathbf{\Phi}$ is a 3×3 matrix of coefficients. This is a standard approach to describe the co-movements of each series; it is parsimonious and allows us to capture the mean-reversion of yields while exploiting the strong cross-correlations between maturities.

Model 3D – Independent GPs

Finally, we test the performance of the GP regression for treasury yields where each maturity is treated as an independent GP regression:

$$Y_{d,t} = f_d(\mathbf{X}_t) + \epsilon_{i,t}, \quad f_d(\mathbf{X}_t) \sim \mathcal{GP}(0, k_d(x,x')), \quad \epsilon_{d,t} \sim \mathcal{N}(0,\sigma_i^2). \tag{31}$$

where the kernel for each output is alike to Model 1E. This allows us to compare whether we are better off modelling different maturities jointly, using fewer latent functions, or whether we should just use one GP for each output.

3.5 Model Training and Evaluation

Each model is initially trained on the 200 months from January 1988 to September 2004. We then fix the hyperparameters and the models are subsequently tested each month by producing 1-step, 3-step, 6-step and 12-step predictions from October 2004 to January 2020. GP models use a rolling 200 point window to update the posterior at each new observation. Each 50 months, the hyperparameters such as kernel lengthscales and coregionalization weights are re-trained so that the models can adapt to new data without too much computational expense. Models with lagged macroeconomic variables are trained using as many lags as they are intended to forecast. This means we always have all the data we need to produce a prediction.

We assess the models' predictions based on their root-mean-squared-error (RMSE) and mean-absolute-error (MAE):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{32}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|. \tag{33}$$

While RMSE and MAE assess the accuracy of point estimates, they do not evaluate the rest of the predictive distribution. As our predictive distributions are assumed to be Gaussian, we can achieve this using a log-scoring rule:

$$L = \log \hat{p}(y_i \mid \hat{y}_i, \hat{\sigma_i}^2) = \tfrac{1}{2}\log(\hat{\sigma_i}^2) + \frac{(y_i - \hat{y}_i)^2}{2\hat{\sigma_i}^2} + \tfrac{1}{2}\log(2\pi). \tag{34}$$

This is the negative log-probability density that the model assigns the true value, meaning lower values are preferred. This rewards forecasts that are both close to the true value and sure in their predictions while penalising over-confidence. If the predictive variance is small but the mean is not close to the true value, the score will be heavily penalised.

Diebold-Mariano Test

Diebold and Mariano (1995) develop a statistical test to compare the accuracy of two sets of forecasts. We first must assign a loss function for our predictions. For our purposes, we can choose either the squared error, absolute error and log-score as our loss functions and define the differential as $d = e_1 - e_2$ where e_1 and e_2 are losses for each set of predictions. The null hypothesis is that both sets of forecasts are equally strong, meaning that d has expectation zero.

$$H_0 : \mathbb{E}[d_t] = 0 \quad \text{vs.} \quad H_1 : \mathbb{E}[d_t] \neq 0. \tag{35}$$

The DM test statistic is

$$DM = \frac{\bar{d}}{\sqrt{\frac{1}{T}\gamma_0 + 2\sum_{k=1}^{h-1}\gamma_k}}, \tag{36}$$

where $\bar{d}$ is the sample mean loss differential and γ_k are the sample autocovariances of d at lag k. Diebold and Mariano show that under H_0, stationarity and ergodicity of d, the DM test statistic converges in distribution to $\mathcal{N}(0,1)$, thus allowing us to compare the predictions of different models with a hypothesis testing approach that is robust to the dependence of time series data.

4 Results and Discussion

Each model is trained on monthly data from January 1988 to September 2004 and subsequently produces 1, 3, 6 and 12 month forecasts for each of the 1, 2, 3, 4, 5, 7 and 10 year US Treasury yields from October 2004 to January 2020. All computation is performed in Python using the `GPy` module. Most models perform similarly in point estimates across the one-step horizon. However, models that handle uncertainty well at short lags (i.e. have high log-scores) often seem to outperform others in RMSE and MAE when forecasting at higher lags. The main results listed below in Table 2 are the average log-score, RMSE and MAE for each model averaged across all maturities.

The Brownian kernel and OU kernel alone narrowly outperform the standard VAR(1) on 3 PCs (Model 3C) approach in log-score across the one-step forecasts. As the forecast horizon lengthens, the natural uncertainty quantification of a GP model starts to improve predictions compared to baseline linear models. That being said, Models 1C and 2A show that if a kernel is poorly specified, a GP model can easily start to struggle.

In one-step log-score, the strongest kernel structure has each latent prior following an OU plus locally periodic structure in time, an RBF kernel in federal funds rate and CPI inflation, and a Brownian kernel for past volatility. The added non-stationarity from the Brownian kernel seems to capture volatility clustering well and allows the model to assign higher variance in periods following high volatility. In fact, Model 3D achieves the best log-score at one lag despite not addressing any correlations between outputs. However, this result is not statistically significantly stronger than the other models that include volatility and when tested on higher lags, does not particularly stand out. Also, this model is not entirely independent across outputs because although the target variables are treated as such, through the exponentially weighted volatility, each output still receives the trace of the volatility across all outputs. While past volatility helps predictions at one lag, it does not outperform Model 1D for 6-step and 12-step predictions, suggesting that past volatility is only useful in the short term.

Model ID	Description
1A	BM
1B	OU
1C	OU + LP
1D	OU + LP + Federal funds, CPI
1E	OU + LP + Volatility + Federal funds, CPI
1F	OU + LP + Volatility + Macros
2A	LP + OU
2B	LP + OU + Federal funds,CPI
2C	LP + OU + Macros
3A	Random walk
3B	ARIMA
3C	VAR(1) on 3 PCs
3D	Independent GPs

Table 1: Key for model identifiers used in the performance results table below.

Model	1 lag			3 lags			6 lags			12 lags		
	Avg LPS	RMSE	MAE	Avg LPS	RMSE	MAE	Avg LPS	RMSE	MAE	Avg LPS	RMSE	MAE
1A	−0.036	0.227	0.162	0.539	0.411	0.294	0.926	0.611	0.435	1.241	0.816	0.604
1B	−0.036	0.226	0.162	0.534	0.408	0.295	0.908	0.599	0.430	1.194	0.773	0.569
1C	−0.003	0.236	0.178	0.622	0.450	0.339	1.044	0.685	0.529	1.560	1.032	0.855
1D	−0.079	0.226	0.161	0.486	0.403	0.301	**0.724**	**0.510**	**0.397**	**0.922**	**0.600**	**0.472**
1E	−0.106	0.221	0.160	0.435	0.386	0.298	0.876	0.621	0.481	0.962	0.621	0.481
1F	−0.044	0.362	0.199	**0.361**	**0.368**	**0.268**	0.949	0.644	0.488	1.427	0.932	0.754
2A	0.033	0.246	0.188	0.569	0.433	0.315	1.047	0.711	0.528	1.236	0.840	0.656
2B	−0.075	**0.218**	**0.157**	0.689	0.484	0.363	0.797	0.519	0.388	1.084	0.623	0.503
2C	0.002	0.241	0.183	0.620	0.519	0.359	0.987	0.658	0.484	1.082	0.687	0.557
3A	−0.029	0.227	0.162	0.545	0.411	0.295	0.928	0.610	0.436	1.256	0.836	0.628
3B	−0.032	0.228	0.162	0.578	0.408	0.291	0.969	0.609	0.436	1.300	0.831	0.625
3C	−0.030	0.235	0.169	0.595	0.446	0.332	1.039	0.677	0.508	1.413	0.932	0.714
3D	**-0.124**	0.227	0.160	0.436	0.407	0.287	0.777	0.564	0.416	1.020	0.672	0.526

Table 2: Model performance (Average log-score, RMSE, MAE) across different lag lengths. Entries in bold represent the strongest score in that metric.

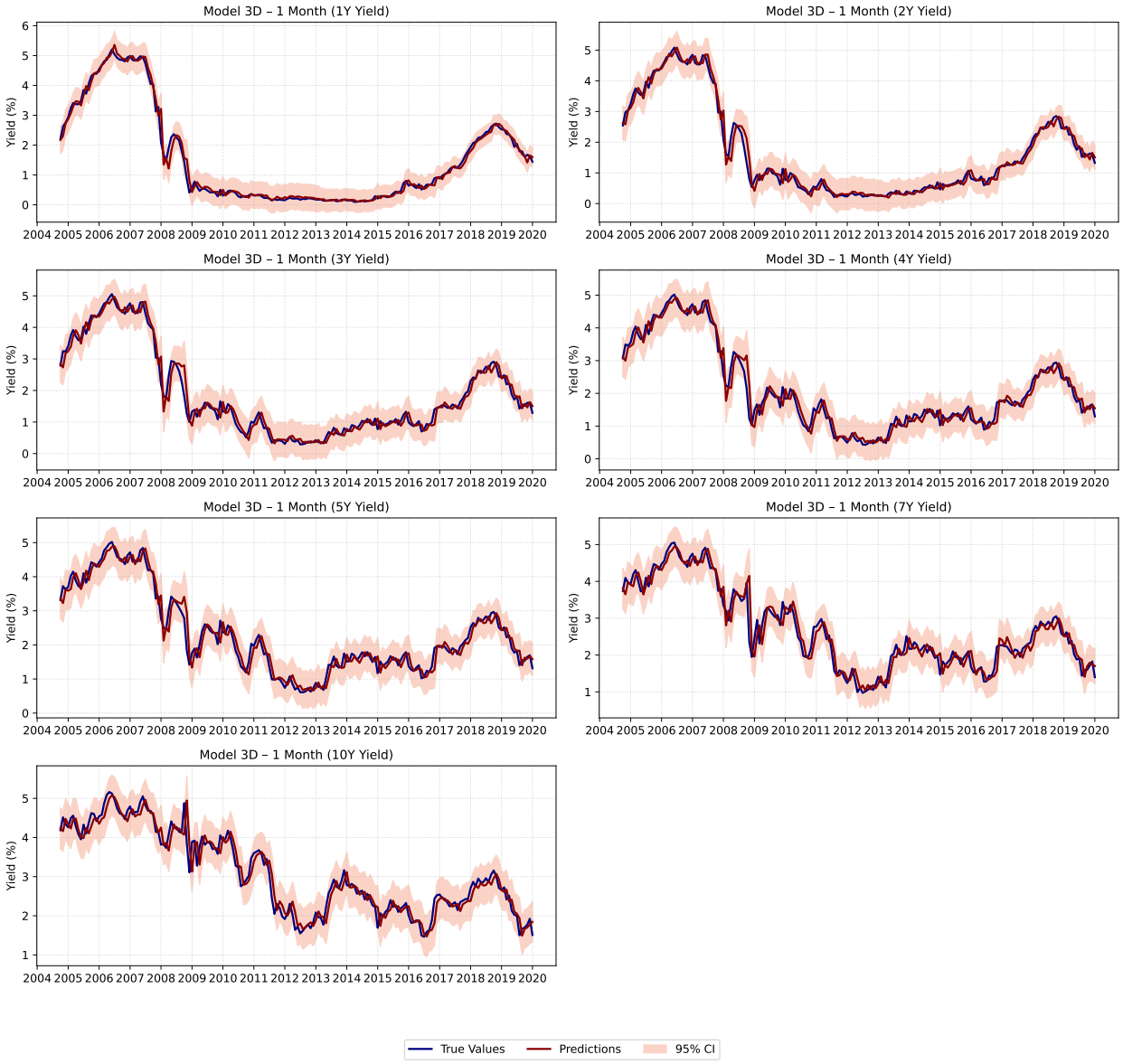


Figure 11: Model 3D (Independent GPs) - 1 month forecasts.

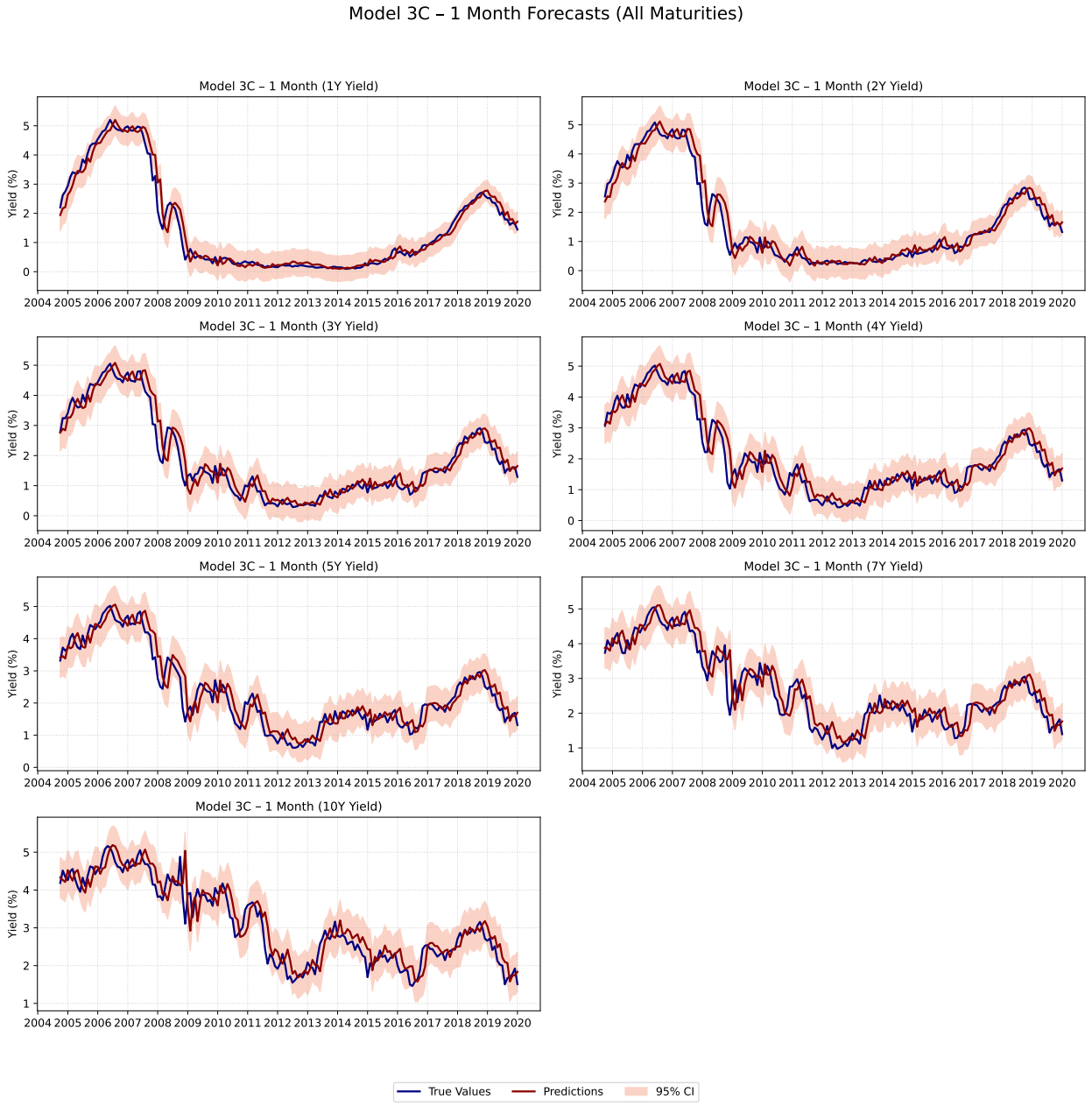


Figure 12: Model 3C (VAR on PCs) - 1 month forecasts.

For three-month predictions, Model 1F stands out across all three metrics, but only at this forecast length. This suggests that the addition of the broad macroeconomic factors helps to describe yield movement in the medium term but can overcomplicate things across other horizons.

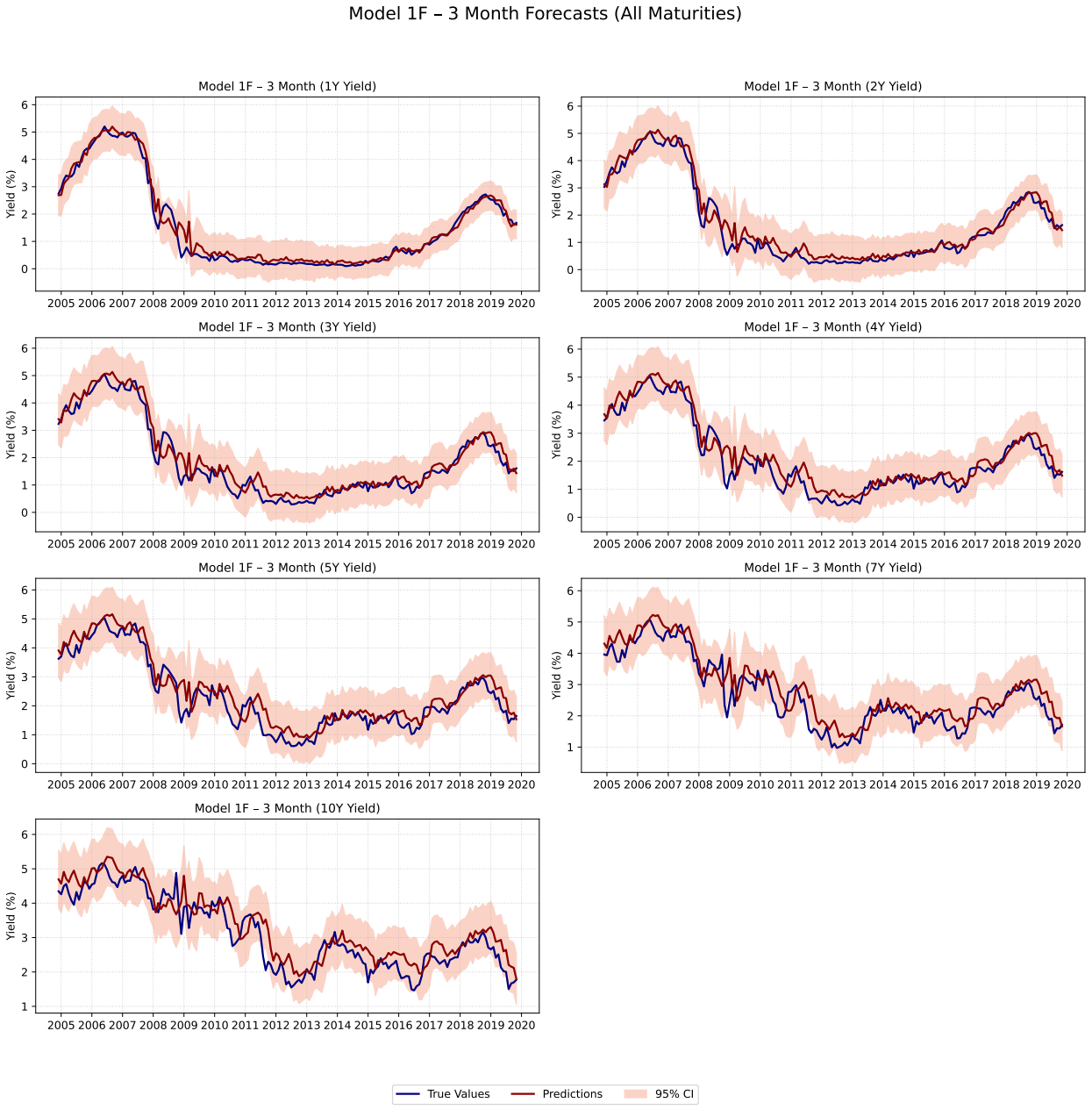


Figure 13: Model 1F (Macro factors) - 3 month forecasts.

The model without volatility but still with federal funds rate and CPI (Model 1D) performs well in six- and twelve-month predictions suggesting that time- and inflation-based metrics may be sufficient in the longer term. It has the best log-score, RMSE and MAE at six and twelve months, though is not statistically significantly better than models that include volatility, according to the DM test. It is important to note that although the addition of the locally periodic structure does not improve performance on its own, it outperforms just an OU kernel when we add covariates, especially at higher lags.

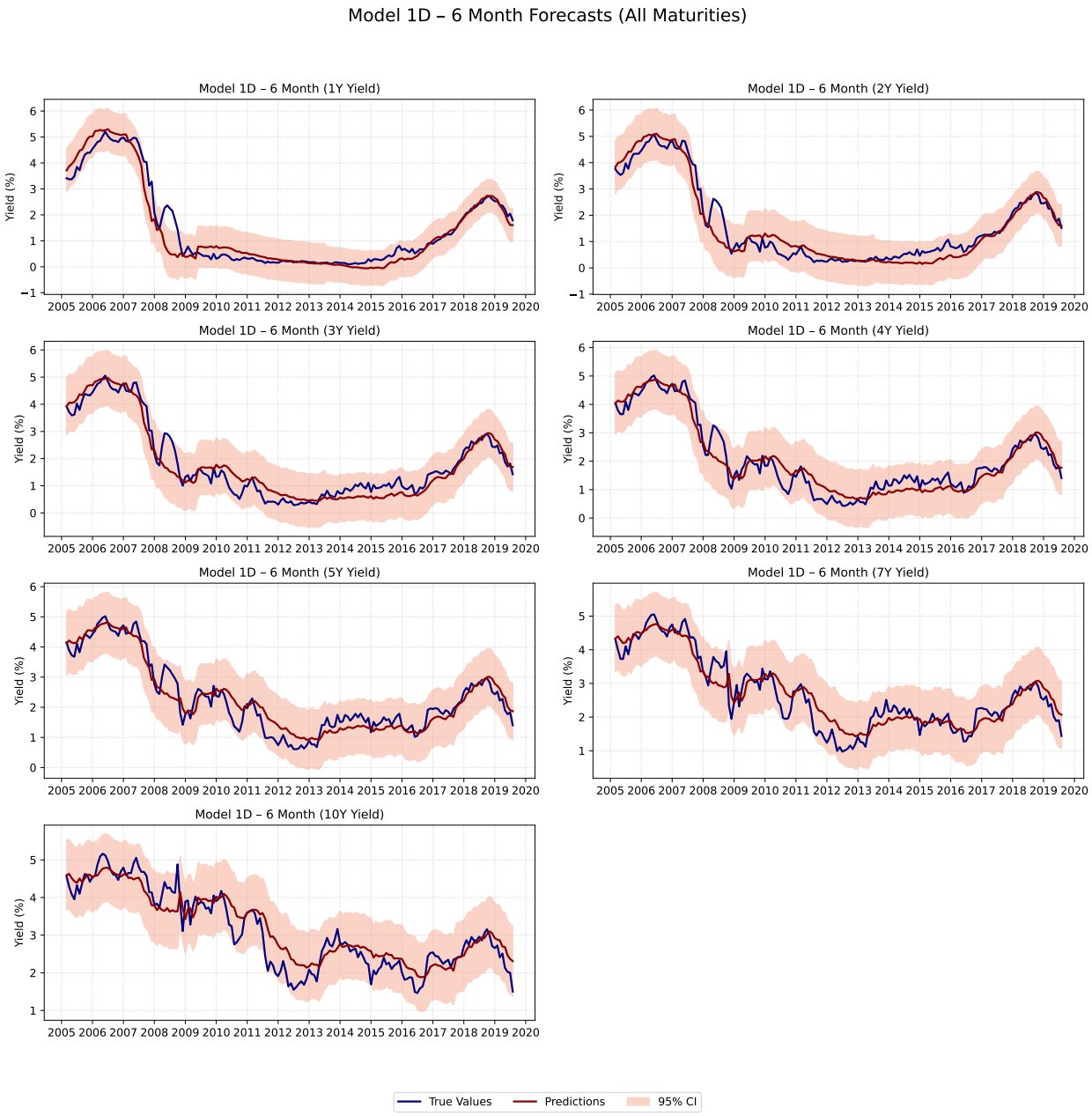


Figure 14: Model 1D (federal funds and CPI) - 6 month forecasts.

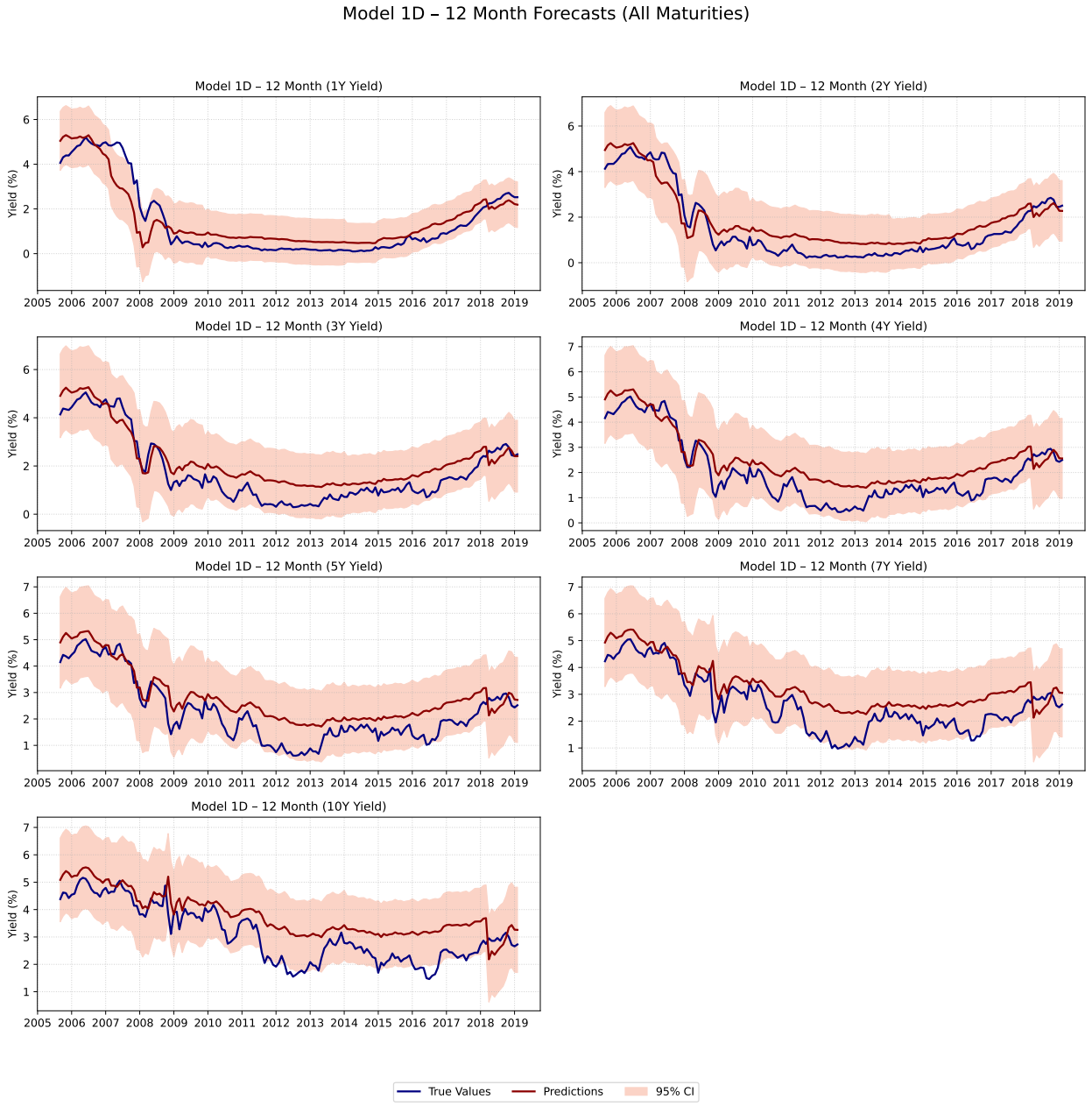


Figure 15: Model 1D (federal funds and CPI) - 12 month forecasts.

Likely due to the strong correlations between maturities, the IMC assumption of one kernel group seems to be enough to describe their shared dynamics. Although the difference is not large, the IMC Model 1D performed slightly better than the equivalent LMC Model 2B, suggesting that letting maturities' weights differ between kernel structures is unnecessary.

Without including any way to explicitly allow long-term expectations to move Treasury yields, models found shorter maturities (one and two year) much easier to forecast than longer maturities, especially seven- and ten-year bonds. In general, GP models maintained this behavior at all forecast lengths but as shown in Table 3, the performance of the VAR

approach inverts. Here, shorter maturities are easier to forecast one month ahead but harder to forecast twelve months ahead. Notably, Model 3D maintains a better grasp on longer maturities than other models, implying that the shared latent functions of correlated models could be fitting more to the short maturities and neglecting the longer maturities. The log marginal likelihood does contain a term to penalise complexity, but the term that encourages goodness-of-fit can often dominate; it will give less attention to fitting some outputs if it can push up the model evidence in others more easily. That being said, the random walk benchmark also finds shorter maturities easier to describe, meaning this effect is likely not solely due to the way that GPs train.

Maturity	Model	Lags	Avg LPS	RMSE	MAE
1 Year	1D	1	−0.414	0.164	0.101
		12	0.887	0.583	0.489
	1E	1	−0.394	0.158	0.101
		12	0.689	0.477	0.390
	3C	1	−0.305	0.171	0.112
		12	1.506	1.005	0.684
2 Year	1D	1	−0.225	0.198	0.134
		12	1.103	0.729	0.522
	1E	1	−0.200	0.187	0.134
		12	0.893	0.578	0.474
	3C	1	−0.109	0.216	0.146
		12	1.495	0.981	0.694
3 Year	1D	1	−0.128	0.220	0.157
		12	1.065	0.701	0.535
	1E	1	−0.108	0.206	0.155
		12	0.991	0.639	0.521
	3C	1	−0.039	0.229	0.167
		12	1.437	0.942	0.718
4 Year	1D	1	−0.060	0.239	0.175
		12	1.069	0.703	0.543
	1E	1	−0.033	0.230	0.173
		12	1.010	0.658	0.532
	3C	1	0.016	0.242	0.179
		12	1.397	0.923	0.729
5 Year	1D	1	−0.051	0.238	0.179
		12	1.082	0.716	0.576
	1E	1	−0.023	0.232	0.179
		12	1.048	0.681	0.538
	3C	1	0.026	0.244	0.184
		12	1.364	0.904	0.730
7 Year	1D	1	0.007	0.263	0.191
		12	0.966	0.623	0.499
	1E	1	0.059	0.260	0.193
		12	1.069	0.697	0.531
	3C	1	0.099	0.267	0.199
		12	1.341	0.890	0.727
10 Year	1D	1	0.004	0.247	0.182
		12	0.965	0.633	0.516
	1E	1	0.030	0.251	0.188
		12	1.092	0.700	0.516
	3C	1	0.103	0.266	0.198
		12	1.351	0.878	0.713

Table 3: Performance comparison of models 1D, 1E, and 3C across different maturities for 1-step and 12-step forecasts.

4.1 Parameter Estimates

In general, parameters for the time kernel vary across models and across lags. Although the time input is the same for different forecast horizons, the lag on the macroeconomic variables changes, therefore altering the inputs and the shape of the likelihood function. A long lengthscale parameter corresponds to a slow rate of decay in correlation between points and a short lengthscale implies autocorrelation decays quickly. Models assign higher lengthscales to the locally periodic part and a shorter lengthscale for the OU part. As intended, the locally periodic part picks up longer term trends and the OU part describes short-term fluctuations. In Model 1D, the variance of the locally periodic part surprisingly shrinks to near zero at 12 lags, suggesting that the model almost ignores this trend component. Despite this, Model 1D outperforms the other models at 12 lags and significantly outperforms the same model with the locally periodic part omitted.

Federal funds rate and CPI both tend to have long lengthscales but CPI is often significantly longer. For example, in Model 1E at three lags, federal funds rate has a lengthscale of 9.26 and CPI inflation has a lengthscale of 21.2. This suggests the effect of the federal funds rate three months ago is felt more than the effect of CPI inflation three months ago; however, both are still useful in forecasting yields. The model with Ludvigsson & Ng (2009) macroeconomic factors performs best at the three-month horizon. In particular, the seventh and ninth factors have lengthscales of 4.35 and 0.565 suggesting yields are more sensitive to changes in these factors than others with much larger lengthscales. Appendix A contains optimised kernel parameters values for selected models.

Through the magnitude of the coregionalization weights of Model 2B, we can observe that while each maturity is similarly weighted in the function for time, shorter maturities are much more sensitive to economic effects than longer maturities. One year yields have a weight of -44 in the macroeconomic latent function while ten-year yields have a weight of only -3. While the sign of the weights is not identified and therefore arbitrary, the absolute magnitudes here tell us that federal funds rate and CPI inflation are more useful in forecasting shorter maturities than longer ones.

The effect of past volatility is subtle yet distinct as Model 1E estimates the Brownian variance parameter at 0.009 per unit of volatility. This suggests the model is adapting uncertainty accordingly when volatility is high but does not overstate the risk, as reflected in the improved average log-score.

4.2 Robustness Checks

To avoid overfitting, priors can be placed on the error variances in a way that discourages them from shrinking to near zero. We choose Gamma priors for support $[0, \infty)$ and specify hyperparameters that encourage a sensible mean and a variance with a suitable amount of flexibility. Instead of maximizing the marginal likelihood, parameters are estimated through *maximum a posteriori* (MAP) so that they solve the MAP optimisation. This finds the parameter values at the posterior mode and provides a way for us to maximise the model evidence subject to our prior beliefs about parameter values. Despite trying multiple different

values and distributions, priors only seem to hurt performance in- and out-of-sample while not showing much benefit in regularisation.

The addition of the locally periodic kernel initially hurt forecasting performance across each metric. However, once we include macroeconomic variables, the locally periodic structure helps goodness-of-fit and improves predictions, especially at higher lags. For this reason, the locally periodic kernel is retained in Models 1D onwards.

The RBF kernel was chosen to model the relationship between yields and macroeconomic variables because of its smoothness and flexibility. If we instead use a linear kernel, predictions worsen across forecast horizons. This shows the benefit of GP models where yields are not constrained to depend on macros linearly.

The choice to retrain GP models every 50 months is made to limit computational complexity and ensure models are adapting throughout time. The value of 50 is chosen to ensure models train at different points in the year but not too often. Having tried retraining only when predictive accuracy worsens, we observe that this does not present any benefits over simply retraining on a fixed interval (see Appendix B).

An alternative approach to including macroeconomic variables is to include them as additional outputs for the model. This eliminates the issue of having to lag variables but imposes that yields and macroeconomic variables depend on each other linearly. We find that this approach does not improve our predictions which is why we instead opt for inclusion as lagged covariates (see Appendix C).

4.3 Limitations

An important limitation of this approach is the issue of practical non-identifiability, or that multiple different parameter values can achieve similar results. Common in kernel-based supervised learning problems, non-convexity in the negative marginal likelihood makes it difficult to find the global minimum. The optimiser can easily get stuck in ridges or flat sections of the marginal likelihood and once it does converge, we have little way to tell if we have the optimal parameter estimates. We also show in Appendix D that coregionalization weights are not identifiable because the weights and latent functions can be rotated by any orthogonal matrix to produce the same posterior. While our predictions are unchanged, this means that the coregionalization weights can easily vary across models, restricting our interpretation of their values.

A strength of many ATSMs is their ability to restrict arbitrage. While the coregionalized GP framework provides a unified way to model bond yield movements, it does not prevent arbitrage. As shown in Figure 16, residuals are cross-correlated across maturities suggesting the assumption of independent Gaussian noise for each output is violated. As the absence of arbitrage is an important restriction governing how maturities move together, further research implementing this could be beneficial. Alternatively, this issue could be tackled by implementing a low-rank approximation for the noise covariance matrix and relaxing the diagonal independence assumption.

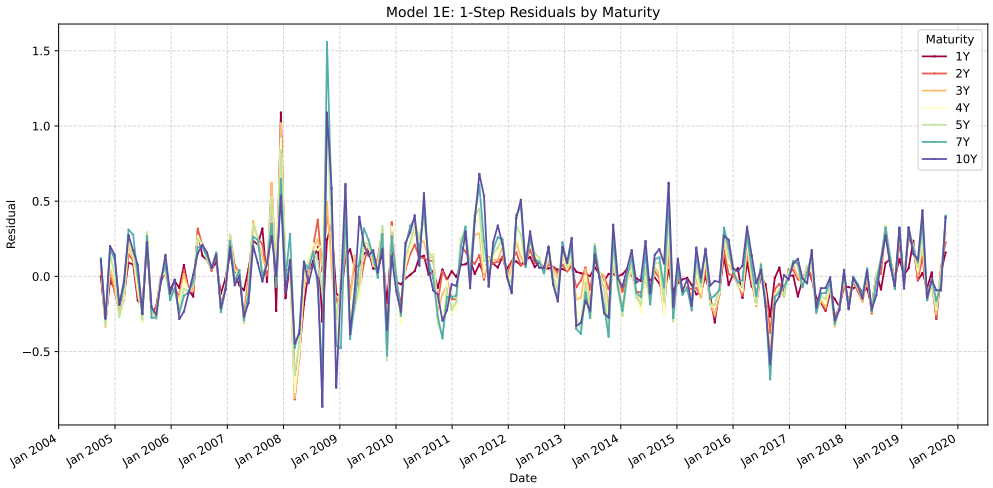


Figure 16: Model 1E - 1 Month forecasting errors by maturity.

As shown above, volatility clustering does not appear to be entirely accounted for, especially around the global financial crisis of 2008. One approach to model heteroscedasticity follows Le et al. (2005), where separate GPs are assigned for the mean and variance of the outputs, which are both input dependent. However, extending this to a multi-output problem is non-trivial and brings additional concerns regarding whether monthly yield data is large enough to avoid overfitting.

5 Conclusion

In this paper, we present a non-parametric Bayesian approach for modelling term structures that is unified, allows for non-linearities and presents benefits out-of-sample compared to traditional VAR approaches. Multi-output GP regression allows us to reduce model complexity through its use of fewer latent functions than outputs and a method of regularisation through maximizing the model evidence. It also offers an avenue for probabilistic predictions, allowing us to quantify our uncertainty and assess models based on their entire predictive distributions.

We find that an exponentially weighted moving average of past volatility is useful in quantifying uncertainty, especially in the short term. We also find that a broad range of macroeconomic factors are beneficial for predicting three months ahead but only the federal funds rate and CPI inflation are needed when forecasting in the longer term. While independent GP regressions perform best in one-month predictions, we find allowing for correlated outputs has clear benefits when forecasting further into the future.

The decay parameter for past volatility is selected separately from the rest of the model. Sensitivity analysis on how this affects forecasts and cross-validation to find the optimal value is an avenue for future work. Furthermore, the restriction of arbitrage is a clear direction for further research that could be beneficial in determining the co-movement of yields across different maturities.

Nevertheless, we show that Gaussian process regression is a strong alternative to observable factor models and presents benefits out-of-sample, especially in quantifying uncertainty. This makes GPs a suitable choice for policy makers or risk managers, providing more accurate estimation for the distribution of future yields.

A Parameter Estimates

Parameter	Value
σ^2_{LP}	0.0145
p	5.9151
ℓ_{per}	3.0101
ℓ_{rbf_1}	11.6401
σ^2_{OU}	0.0029
ℓ_{OU}	0.6432
$\sigma^2_{\text{rbf}_2}$	0.1603
$\ell_{\text{rbf}_{2,fed}}$	8.9922
$\ell_{\text{rbf}_{2,cpi}}$	19.0904
$\sigma^2_{\text{Brownian}}$	0.0076

Table 4: Kernel parameter values for Model 1E at 1-step forecast horizon.

Parameter	Value
σ^2_{LP}	0.0009
p	2.9570
ℓ_{per}	1.2972
ℓ_{rbf_1}	1.8297
σ^2_{OU}	0.0002
ℓ_{OU}	0.1796
$\sigma^2_{\text{rbf}_2}$	0.0243
$\ell_{\text{rbf}_{2,cpi}}$	8.2434
$\ell_{\text{rbf}_{2,fed}}$	142.1357
$\ell_{\text{rbf}_{2,1}}$	118.5814
$\ell_{\text{rbf}_{2,2}}$	79.2445
$\ell_{\text{rbf}_{2,3}}$	167.7034
$\ell_{\text{rbf}_{2,4}}$	27.4204
$\ell_{\text{rbf}_{2,5}}$	62.0868
$\ell_{\text{rbf}_{2,6}}$	124.4266
$\ell_{\text{rbf}_{2,7}}$	4.6480
$\ell_{\text{rbf}_{2,8}}$	127.8055
$\ell_{\text{rbf}_{2,9}}$	0.5350
σ^2_{BM}	0.0002

Table 5: Kernel parameter values for Model 1F at the 3-step forecast horizon.

Parameter	Value	
	6 Lags	12 Lags
σ^2_{LP}	0.0093	0.0000
p	7.2098	13.5359
ℓ_{per}	1.0391	6.6297
ℓ_{rbf_1}	5.0745	10.3476
σ^2_{OU}	0.0023	0.0255
ℓ_{OU}	0.2526	3.0928
$\sigma^2_{\text{rbf}_2}$	0.1304	0.0719
$\ell_{\text{rbf}_{2,fed}}$	6.5619	1.3542
$\ell_{\text{rbf}_{2,cpi}}$	160.3654	54.7003

Table 6: Kernel parameter values for Model 1D at the 6-step and 12-step forecast horizons.

B Retraining Frequency

Retraining Strategy	Avg LPS	RMSE	MAE
Every 50 months	−0.106	0.221	0.160
When non-stationary (ADF $p > 0.1$)	−0.077	0.219	0.160

Table 7: Performance comparison of Model 1E with different retraining strategies: Retraining every 50 months versus retraining based on non-stationarity detection (Augmented Dickey-Fuller p-value > 0.1).

C Macroeconomic Covariates as Outputs

VAR Model Specification	Avg LPS	RMSE	MAE
Without LN Factors	−0.0298	0.2350	0.1690
With LN Factors	−0.0133	0.2411	0.1733

Table 8: Performance comparison of VAR models with and without Ludvigsson & Ng (2009) macroeconomic factors at the 1-step horizon. The inclusion of macroeconomic factors does not improve predictions.

D Non-Identifiability of Coregionalization Parameters

The covariance for a multi-output GP under the Linear Model of Coregionalization (LMC) is:

$$\mathbf{K}(\mathbf{x}, \mathbf{x}') = \sum_{q=1}^{Q} \mathbf{B}_q \otimes k_q(\mathbf{x}, \mathbf{x}') \tag{37}$$

Let $\mathbf{Q} \in \mathbb{R}^{D \times D}$ be an orthogonal matrix ($\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). Define a rotated set of coregionalization matrices:

$$\tilde{\mathbf{B}}_q = \mathbf{Q}^\top \mathbf{B}_q \mathbf{Q} \tag{38}$$

The resulting covariance function $\tilde{\mathbf{K}}$ is:

$$\begin{aligned}
\tilde{\mathbf{K}}(\mathbf{x}, \mathbf{x}') &= \sum_{q=1}^{Q} \tilde{\mathbf{B}}_q \otimes k_q(\mathbf{x}, \mathbf{x}') \\
&= \sum_{q=1}^{Q} (\mathbf{Q}^\top \mathbf{B}_q \mathbf{Q}) \otimes k_q(\mathbf{x}, \mathbf{x}') \\
&= (\mathbf{Q}^\top \otimes \mathbf{I}) \left(\sum_{q=1}^{Q} \mathbf{B}_q \otimes k_q(\mathbf{x}, \mathbf{x}') \right) (\mathbf{Q} \otimes \mathbf{I}) \\
&= (\mathbf{Q}^\top \otimes \mathbf{I})\, \mathbf{K}(\mathbf{x}, \mathbf{x}')\, (\mathbf{Q} \otimes \mathbf{I})
\end{aligned} \tag{39}$$

While $\mathbf{K} \neq \tilde{\mathbf{K}}$, the model is observationally equivalent. A vector $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$ rotated by $(\mathbf{Q} \otimes \mathbf{I})$ has the distribution:

$$\tilde{\mathbf{f}} = (\mathbf{Q} \otimes \mathbf{I})\mathbf{f} \sim \mathcal{N}\left(\mathbf{0}, (\mathbf{Q} \otimes \mathbf{I})\mathbf{K}(\mathbf{Q}^\top \otimes \mathbf{I})\right) = \mathcal{N}(\mathbf{0}, \tilde{\mathbf{K}}) \tag{40}$$

As the rotation is invertible, the probability distribution of the data $\mathbf{y} = \mathbf{f} + \boldsymbol{\epsilon}$ is identical under both parametrisations. The likelihoods $\mathcal{N}(\mathbf{y}|\mathbf{0}, \mathbf{K} + \sigma_n^2 \mathbf{I})$ and $\mathcal{N}(\mathbf{y}|\mathbf{0}, \tilde{\mathbf{K}} + \sigma_n^2 \mathbf{I})$ are therefore equivalent, and the parameters $\mathbf{B}_q$ cannot be uniquely identified from the data. While this does not affect our predictions, this does affect our interpretation of parameter values.

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